UGANDA CHRISTIAN UNIVERSITY

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FACULTY OF ENGINEERING, DESIGN AND TECHNOLOGY

DEPARTMENT OF COMPUTING AND TECHNOLOGY

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Course Unit: Computer Organization and Architecture
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DATA REPRESENTATION

Section A: Number Base Conversions

Part 1: Basic Conversions

1(a) 101101_2

$(1\times2^5)+(0\times2^4)+(1\times2^3)+(1\times2^2)+(0\times2^1)+(1\times2^0)$

$=45_{10}$

(b) 110010_2

$=50_{10}$

(c) 100111_2

$=39_{10}$

2 (a) 27_{10}

2	27	1
2	13	0
2	6	0
2	3	1
	1	

$=11011_2$

(b) 45_{10}

2	45	1
2	22	0
2	11	1
2	5	1
2	2	0
	1	

$=101101_2$

(c) 64_{10}

2	64	0
2	32	0
2	16	0
2	8	0
2	4	0
2	2	0
	1	

$\underline{\underline{1000000_2}}$

3(a) $27_8 = (2\times8^1)+(7\times8^0)$

$= \underline{\underline{23_{10}}}$

$45_8 = (4\times8^1)+(5\times8^0)$

$= \underline{\underline{37_{10}}}$

$63_8 = (6\times8^1)+(3\times8^0)$

$= \underline{\underline{51_{10}}}$

4(a) 25_{10}

8	25	1
	3	

$= \underline{\underline{31_8}}$

(b) 72_{10}

8	72	0
8	9	1
	1	

$= \underline{\underline{110_8}}$

(c) 90_{10}

8	90	2
8	11	3
	1	

$= 132_8$

Part 2: Hexadecimal Conversions

5 (a) $3A_{16} = (3 \times 16^1) + (10 \times 16^0)$

$= 58_{10}$

(b) $7F_{16} = (7 \times 16^1) + (15 \times 16^0)$

$= 127_{10}$

c) $B4_{16} = (11 \times 16^1) + (4 \times 16^0)$

$= 180_{10}$

6(a) 50_{10} $\cancel{16}$

16	50	2
	3	

$= 32_{16}$

(b) 75_{10}

16	75	11(B)
	4	

$= 4B_{16}$

(c) 120_{10}

16	120	8
	7	

$= 78_{16}$

Part 3: Cross Base Conversions

7 (a) 110111_2 $= 67_8$

(b) 101101_2 = 101 = 5

~~101 = 5~~

$= 55_8$

8 (a) 11101011_2

$= EB_{16}$

(b) 10101101_2

$= AD_{16}$

9 (a) $5C_{16}$

5	0101
C	1100

$= 01011100_2$

(b) $A7_{16}$

A	1010
7	0111

$= 10100111_2$

Section B: Fractional Conversions

10 (a) $101.101_2 = (1\times2^2)+(0\times2^1)+(1\times2^0)+(1\times2^{-1})+(0\times2^{-2})+(1\times2^{-3})$

$= 5.625_{10}$

(b) $11.011_2 = (1\times2^1)+(1\times2^0)+(0\times2^{-1})+(0\times2^{-2})+(0\times2^{-3})$

$= 3.375_{10}$

11 (a) 10.25_{10} Whole

2	10	0
2	5	1
2	2	0
	1	

Fractional

2×0.25	0
2×0.5	1

$= 1010.10_2$

(b) 7.5_{10} Whole

2	7	1
2	3	1
	1	

Fraction

0.5×2	1

$= 111.1_2$

12(a) 26.5_{10} Whole

16	26	10 (A)
	1	

Fraction

0.5×16	8

$= 1.A_{16}$

(b) 15.75_{10} Whole

16	15	15(F)
	0	

Fraction

0.75×16	12(C)

$= F.C_{16}$

Section C: ASCII / EBCDIC Character Conversions

13.

Character	ASCII decimal
A	65
a	97
9	57
#	35

15.

Character	ASCII	Binary
C	67	01000011
F	70	01000110
7	55	00110111

14.

ASCII code	Character
65	A
97	a
48	0
36	$

~~15.~~

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16.

D	68
A	65
T	84
A	65

17.

72	H
69	E
76	L
76	L
79	O

The word is HELLO

~~18~~.

18. EBCDIC was developed by IBM and was used mainly on IBM mainframe systems while ASCII was developed by ANSI and it is used on most modern computers.
Characters have different numeric codes for example: A in ASCII is 65 while in EBCDIC it's 193.

19. Computer hardware is made of transistors that are only either on or off. This on or off can be labelled as 1 or 0. Decimal representation would require ten different voltages which is power consuming and less reliable on the other end, binary keeps circuits simple, fast and effective.

20. One hexadecimal digit represents ~~[illegible]~~ four binary ~~dig~~ bits which means long binary strings can be written in ~~a~~ much shorter format. An example would be an 8 bit byte only needing 2 hexadecimal digits. This makes memory addresses and raw data easier for programmers to read and check than binary.